



Figure 1: ML.NET attributes

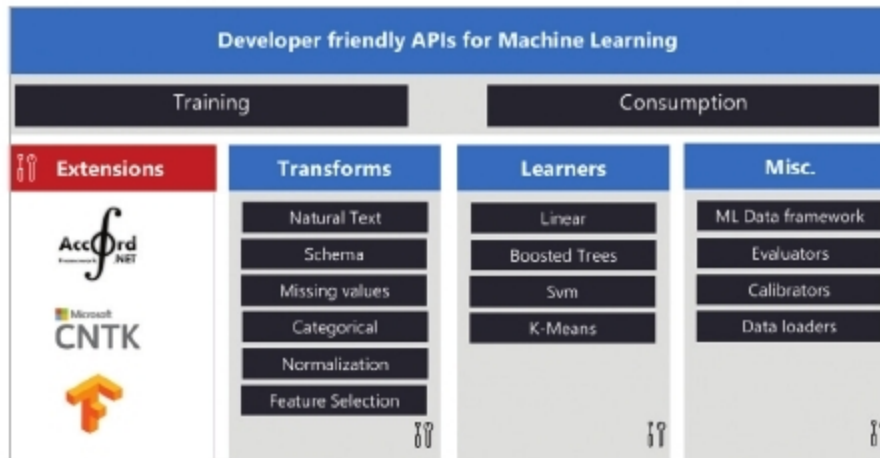


Figure 2: ML.NET components

- The power of ML.NET has been proven with popular Microsoft features such as Windows Hello, Bing Ads.
- It has support for various ML scenarios such as sentiment analysis and recommendations. It supports deep learning (DL) scenarios also, such as image classification.

The complete list of ML.NET's components is shown in Figure 2 (Source: <https://blogs.msdn.microsoft.com/dotnet/2018/05/07/introducing-ml-net-cross-platform-proven-and-open-source-machine-learning-framework/>).

ML.NET has components that support all aspects of ML, which include core data types, customisable pipelines, high performance math, data structures for heterogeneous data, tooling support, etc.

Installing and building your first app with ML.NET

As stated earlier, the ML.NET framework is cross-platform. It can be installed in the Windows environment with the following steps.

- **Step 1:** Download and install the .NET software development kit (.NET SDK).
- **Step 2:** Create the .NET app from the command prompt with the following commands:

```
> dotnet new console -o myApp
> cd myApp
```

The above command builds a command line based .NET app in the directory named *myApp* and fills it with the required files for the app.

- **Step 3:** Install the ML.NET package by executing the following command:

```
> dotnet add package Microsoft.ML --version 0.4.0
```

For the Ubuntu 18.04 environment, the installation

requires the following commands. First register the Microsoft key and the product repository, and then install the required dependencies:

```
wget -q https://packages.microsoft.com/config/ubuntu/18.04/
packages- microsoft-prod.deb
sudo dpkg -i packages-microsoft-prod.deb
```

Next, install the .NET SDK as shown below:

```
sudo apt-get install apt-transport-https
sudo apt-get update
sudo apt-get install dotnet-sdk-2.1
```

Download the ML.NET package with the following code:

```
$ dotnet add package Microsoft.ML
```

- **Step 4:** Any ML application has two components — the data set and the algorithm used to perform the actual task. For an ML app to succeed, both these components should be powerful. In this example application, we are going to work with a popular data set used by learners of ML applications – the iris data set. The purpose of this data set is to predict the type of iris flower. The iris data set can be downloaded from <https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data>. In Visual Studio, this data set can be copied on to the app's output directory.
- **Step 5:** The *Program.cs* file is the core file containing the code. This file starts with the inclusion of libraries (Code: <https://www.microsoft.com/net/learn/machinelearning-ai/ml-dotnet-get-started-tutorial/>):

```
using Microsoft.ML;
using Microsoft.ML.Data;
using Microsoft.ML.Runtime.Api;
using Microsoft.ML.Trainers;
using Microsoft.ML.Transforms;
using System;
```

Indicate the input features and class labels with the following code:

```
public class IrisData
{
    [Column("0")]
    public float SepalLength;

    [Column("1")]
    public float SepalWidth;

    [Column("2")]
    public float PetalLength;
```